# Jueun **Kim**

Ph.D. Student in CS, POSTECH

jekim@dblab.postech.ac.kr | juevn.github.io | Google Scholar

## EDUCATION

**Ph.D.**, Computer Science and Engineering *Sep 2025 – Present*

*Pohang University of Science and Technology (POSTECH), Pohang, South Korea*

- Advisor: Prof. Wook-Shin Han

**B.S.**, Computer Science and Engineering *Feb 2021 – Aug 2025*

*Pohang University of Science and Technology (POSTECH), Pohang, South Korea*

- Ranked 1st in the department
- Graduated Summa Cum Laude
- GPA: 4.04/4.3

**Exchange Student**, Computer Science *Sep 2024 – Feb 2025*

*Technical University of Munich (TUM), Munich, Germany*

- Achieved top grades in Deep Learning and NLP coursework

## RESEARCH INTERESTS

Multi-hop and multi-modal information retrieval; retrieval for task-performing agents; long-context management; agent memory

## EXPERIENCE

**Intern** *Jul 2024 – Aug 2024*

*SK hynix*

- Mobile FTL Team — developed an AI model predicting user commands to improve flash translation layer performance

**Student Intern** *Jun 2023 – Aug 2025*

*Data Systems Lab, POSTECH*

- Research Project I: proposed the Text-DB QA task and built its benchmark (ICLR 2026)
- Research Project II: developed an open-domain text-table QA model

## PUBLICATIONS

**CANOPY: Adaptive-Granularity Evidence Compression for Multimodal RAG** *arXiv preprint arXiv:2610.00923*

Hyojeong Yun*, **Jueun Kim***, Wook-Shin Han (*equal contribution)

**Hi-Q: Hierarchical Evidence-guided Query Refinement for Multi-Hop Question Answering** *NeurIPS 2026*

**Jueun Kim**, Sungho Park, Wook-Shin Han

**SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables** *ICLR 2026*

Sungho Park, **Jueun Kim**, Wook-Shin Han

## PROJECTS

**CANOPY: Adaptive-Granularity Evidence Compression for Multimodal RAG** *2026*

*arXiv preprint arXiv:2610.00923*

Represents retrieved items as hierarchies and selects evidence at several granularities at once, since choosing a retrieval granularity does not settle how much of each item to keep — cut reader-input evidence tokens by 14.2–27.7% at comparable accuracy across five QA benchmarks over a 33M-item heterogeneous corpus.

**Hi-Q: Hierarchical Evidence-guided Query Refinement for Multi-Hop Question Answering** *2026*

*NeurIPS 2026*

Grows a query tree whose shape is decided by whether retrieved evidence already supports each unit, rather than by a fixed decomposition template or a pre-built graph — 52.3 EM / 64.0 F1 averaged over three multi-hop QA benchmarks under full-corpus retrieval, 15.1 EM over IRCoT.

**SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables** *2026*

*ICLR 2026*

Automates table-text multi-hop benchmark construction by unifying tables and text into one relational representation and generating nested queries at a target hop count — thousands of QA pairs at a quarter of HybridQA's annotation time, on which models scoring above 70 F1 on HybridQA lose more than 30 F1.

**Home AI Agent with Event Memory** *2026*

*Korean government-hosted AI competition — finalist (top 20 of 1,100+ teams)*

Merges continuing home-camera scenes into events while preserving observation counts and state transitions, so earlier situations can be recovered without replaying footage — 72 observations into 6 events (91.7% fewer list entries), citing the designated event in 6 of 7 past-event questions.

**Life-Event-Based Public Service AI Agent** *2026*

*Korean government-hosted AI competition — finalist (top 20 of 1,100+ teams)*

Keeps the user's conditions and completion state as persistent memory and uses it to decide what to surface next across procedures spanning multiple steps and deadlines — 806 procedure items across 50 life events linked by 639 precedence relations, cutting precedence violations from 242 under deadline-only ordering to 0 on a 500-case regression.

## TALKS

**SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables** *Jul 2026*

*2026 Global AI Frontier Lab Workshop, Seoul*

## HONORS & AWARDS

**POSTECHIAN Fellowship (Creative)** *Nov 2025*

*POSTECH*

**Grand Prize, SK hynix Internship Program** *Aug 2024*

*SK hynix*

**Popular Award, Undergraduate Research Project II** *Jun 2024*

*POSTECH Computer Science and Engineering Department*

**National Science & Technology Scholarship** *Aug 2023*

*KOSAF (Korea Student Aid Foundation)*

**Global Leadership Program Scholarship** *Sep 2022*

*POSTECH Computer Science and Engineering Department*

## TEACHING

**Teaching Assistant — AI Application Specialist Course** *2026*

*Samsung Electronics*

**Teaching Assistant — Introduction to Data Analysis** *2026*

*POSTECH*

**Lecturer — POSTECHx Lecture, Pintos** *2024*

*POSTECH*

**Teaching Assistant — Data Programming Course** *2023*

*Samsung Electronics*